

Uganda's power sector reform: There and back again?

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ABSTRACT

Uganda occupies a unique space in the history of power sector reform and private electricity investment in Africa. In this article, we describe the drivers for reform as well as the reform process, including the main institutions involved and the legislative and policy instruments that helped shape them. We then offer a brief history of independent power projects (IPPs) in Uganda and emerging Chinese-funded projects, exploring the nexus between the two. Like any other private investment, IPPs require a suitably enabling environment offering long-term investment certainty. Our primary objective is to assess the experience of Uganda's IPPs and consider what might help accelerate private investment in the country's electricity sector. Finally, we attempt to extract the policy lessons that may be learned from this fascinating story.

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Introduction

For several decades, Uganda suffered from inadequate power supply. Civil war (1971–1986) and declining water levels in Lake Victoria, the primary reservoir for Uganda's hydro-based electric power system, made this problem worse, causing a 60% decrease in the country's available generation capacity (of less than 500 MW), and access rates at just 7% (ESMAP, 1999: 11; World Bank, 2017a). To deal with this crisis, starting in the late 1990s, the government embarked on the most ambitious power sector reform program as yet seen in Africa (Kapika and Eberhard, 2013: 85).

Uganda was the first African country to unbundle its generation, transmission, and distribution utilities, and offer private concessions for power generation and distribution. Its 250 MW Bujagali independent power plant (IPP) ranks among the largest privately financed hydroelectric power projects in sub-Saharan Africa, and its 50 MW Tororo-Electro-maxx Thermal plant is often considered the first indigenous African IPP, financed built and operated solely by African companies. Despite concerns that the country's power system was too small to reap the benefits that might flow from introducing competition in generation, and ongoing challenges related to the introduction of more focused management of transmission and distribution, the reforms initiated between 1999 and 2005 are beginning to bear fruit.

Losses are down; collections, investment, connections are up and access rates presently stand at 20% (Kapika & Eberhard, 2013; World Bank, 2017a).

In addition, a raft of smaller and medium-sized plants have been or are in the process of being built, with installed capacity reported as 857.5 MW (ERA, 2017). Included among these plants are several small hydro, biomass, and solar projects implemented under the government's co-operation with the German development bank KfW. Furthermore, additional renewable projects developed by IPPs with an accumulated capacity of up to 150 MW have reached financial close.¹ Alongside these private-sector successes, Uganda has two large Chinese-funded hydropower projects currently under construction, which were expected to start commercial operations from 2017/18, though presently are delayed and will likely be commissioned in 2018 and 2020 respectively (Eberhard, Gratwick, Morella, & Antmann, 2016).

The process has not, however, always been smooth. In 2006, for example, the country faced a severe power supply shortage, necessitating emergency power installations. In 2008, the government and the major distribution utility entered a dispute over their contractual obligations. In 2009, a new energy minister, backed by several parliamentarians, tried to reverse a number of the 1999 reforms, but was eventually convinced to back down. During the same period, anti-fraud investigations, allegedly targeting inflated power prices, led to raids of nearly all power sector institutions (Kapika and Eberhard, 2013: 103). Even now, private investment in the power sector tends to be politically contested. The tedious coordination efforts and transaction costs occasioned by the multitude of financiers involved in the Bujagali plant apparently left a lasting impression on Ugandan government officials, who often see private sector involvement as more costly, complex, and time consuming than government procured projects.

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¹ Megawatt and dollar figures in this article are based on the date of financial close.

Uganda offers considerable experience in relation to power sector reform, private sector participation, competitive bidding, grid-connected renewable energy, and Chinese-supported projects. In this article, we describe the drivers for reform as well as the reform process, including the main institutions involved and the legislative and policy instruments that helped shape them. We then offer a brief history of IPPs in Uganda and Chinese-funded projects, including the nexus between the two. Like any other private investment, IPPs require a suitably enabling environment offering long-term investment certainty. Our primary objective is to assess the experience of Uganda's IPPs and consider what might help accelerate private investment in the country's electricity sector. Finally, we attempt to extract the policy lessons that may be learned from this fascinating story and outline future trajectories for Uganda's power sector development.

Data sources

The IPPs discussed in this article are greenfield, grid-connected installations of approximately 5 MW or more, that have reached financial close, are under construction, or are in operation.² A significant amount of data on these installations was collected and analysed. The World Bank's Private Participation in Infrastructure (PPI) database, AidData, and the [China Africa Research Initiative \(CARI\)](#), 2016 project database were our starting point. In addition, we reviewed annual reports and web sites of Uganda's regulator (ERA), utilities (ESKOM Uganda, UETCL and UMEME) and independent power producers, as well as relevant press reports and news articles. Finally, we consulted stakeholders within each of the relevant organizations (between 2015–2017 in person and via email), including the GETFIT Secretariat, where one of our authors was based at the inception of the research. Information concerning the composition of investments by funding source, the terms of IPP contracts (which remain mostly confidential) and the size, composition, and types of investment from emerging financiers (notably China) was gathered from the abovenoted sources and reconfirmed by at least one additional source. Due to the sensitivity of some data, the names of stakeholders have been excluded. Although an unprecedented amount of data was collected and analysed, certain limitations are noted. Because nearly every Chinese-funded generation project³ is directly negotiated with the relevant government agency, limited public data are available. Additional interviews were conducted by authors, and actual projects were verified with stakeholders in Uganda and abroad.

Drivers for reform: a survey of relevant literature & methodological approach

Virtually all major power generation throughout Africa, including in Uganda, was financed by public coffers, including concessionary loans from development finance institutions (DFIs) at the beginning of the 1990s. Publicly financed generation assets were considered core elements in state-owned, vertically integrated power systems ([Yergin & Stanislaw, 2002](#)). Over the course of the 1990s, however, a range of factors caused this to change. The main drivers were identified as insufficient public funds for new generation and decades of poor performance by state-run utilities ([Bacon, 1995](#); [Besant-Jones, 2006](#); [Jhirad, 1990](#); [Kessides, 2004](#); [Moore & Smith, 1990](#); [Victor & Heller, 2007](#); [Wolak, 1998](#); [World Bank, 1993](#)). Subsequently, African countries

began to adopt a new 'standard' model for their power systems, influenced by pioneering reformers in the US, the UK, Chile and Norway ([Patterson, 1999](#); [World Bank, 2003](#)).

The standard model for power sector reform has been roughly defined as a series of steps that move vertically-integrated utilities towards competition, and generally include the following activities: corporatisation, commercialisation, passage of the requisite legislation, establishment of an independent regulator, introduction of IPPs, restructuring/unbundling, divestiture of generation and distribution assets and introduction of competition ([Adamantiades, Besant-Jones, & Hoskote, 1995](#); [Bacon, 1999](#); [Besant-Jones, 2006](#); [Gratwick & Eberhard, 2008](#); [Williams & Ghanadan, 2006](#)).

IPPs with long-term power purchase agreements (PPA) with the state utility, became a priority within overall power sector reform ([World Bank, 1993](#); [World Bank and USAID, 1994](#)). IPPs were considered a solution to persistent supply constraints, and could also potentially serve to benchmark state-owned supply and gradually introduce competition ([APEC Energy Working Group, 1997](#)). IPPs could be undertaken before sector unbundling. An independent regulator was also not a prerequisite since the PPA laid down a form of regulation by contract.

While IPPs were conceived of as part of a larger power sector reform program, reforms were not far-reaching and IPPs subsequently fit precariously into an imperfect structure ([Woodhouse, 2006](#)). Most state utilities remained vertically integrated and maintained a large share of the generation market, with private power invited only on the margin of the sector.

Policy frameworks and regulatory regimes, necessary to maintain a competitive environment, were limited. International competitive bids for those IPPs that were developed were often not conducted because of tight timeframes, resulting in limited competition for the market and, due to long-term PPAs, no competition in the market ([Kapika & Eberhard, 2013](#); [Malgas & Eberhard, 2011](#)). These long-term PPAs and often government guarantees and security arrangements, such as escrows and liquidity facilities, exposed countries to significant exchange-rate risks.

Although Africa has seen private participation in greenfield electricity projects continue, private investment has been erratic, with a spike in 2008, largely due to the financial close of one large project, Nigeria's Afam VI, followed by a trough and then another flurry of activity from 2012 onward, across the continent ([Eberhard et al., 2016](#)).

The result has been a hybrid (part private, part public) solution. This was neither expected nor predicted by the early framers of power sector reform model and therefore requires fresh analysis and a new approach as stakeholders find themselves in untrodden territory with prominent state-owned and private actors.

Specifically, the Ugandan economy is unable to fund the country's growing power needs, including via local capital markets, which do not offer the financial products for long-term power investments. Since 2000, real GDP growth has averaged 6.9% a year and GDP per capita has multiplied four-fold ([Davies et al., 2016:8](#)). At the same time, it is important to note that demand growth has been constrained mainly due to the pending industrialization of the country, the absence of expected large-demand consumers from the oil industry as well as lagging progress in electrification and low access rates (the consumer profile is roughly: 24% domestic consumers, 12% commercial, 17% medium industry and 46% large industry, with street lighting making up the balance, of a total of the 2,274,544 megawatt hours annually ([MEMD, 2014:11](#)). Furthermore, consumer demand patterns have changed and more consumption occurs in off-peak hours, with Uganda having among the higher electricity rates in Sub-Saharan Africa, with consumers spending approximately 4.5% of household monthly income on electricity ([Kojima and Trimble, 2016: 20, 45](#); [Okoboi & Mawejje, 2016](#)).⁴

² A complete list of IPPs may be found in [Appendix A](#).

³ The typical project structure of most Chinese-funded generation projects (of which there were 34 across Sub-Saharan Africa between 1990 and 2014 totaling 7.5 GW) involves a contractor plus a financing contract. The majority of these projects receive funding from the China ExIm Bank (responsible for soft loans and export credit) on behalf of the Chinese government. Additional finance is provided by other banks (such as the China Development Bank, the Industrial And Commercial Bank of China, the China Construction Bank, and the Bank of China), owned in whole or in part by the Chinese government ([Eberhard et al., 2016](#)).

⁴ In [Kojima and Trimble \(2016\)](#), out of 22 Sub-Saharan African countries, 16 countries rank less expensive than Uganda in terms of monthly bill for 30 kWh.

Uganda's utilities do not have investment-grade ratings so they struggle to raise sufficient debt at affordable rates in international markets. Official development aid and concessionary funding has fluctuated considerably since 2006, but remains available to the Ugandan government and public utilities.⁵ However, since 2012, it's been overshadowed by Chinese and IPP-investment, the latter, being supported by innovative blended financing programs.

The challenge remains daunting: two out of three households in Sub-Saharan Africa, close to 600 million people, have no electricity connection at all. Thus, only 37% of the population currently has electricity access (and just 20% in Uganda), and poor supply is the rule, not the exception. The cost of meeting Sub-Saharan Africa's power sector needs has been estimated at \$40.8 billion a year, equivalent to 6.35% of Africa's GDP (Castellano, Kendall, Nikomarov, & Swemmer, 2015; Eberhard et al., 2011: 149).⁶ Approximately two thirds of the total spending is needed for capital investment (\$26.7 billion a year); the remainder is for operations and maintenance (O&M), which is presently covered mostly by the public sector. Of capital investment, about \$14.4 billion is required for new power generation each year, and the remainder for refurbishments and networks (Eberhard et al., 2011: 149). Existing annual expenditures, of approximately US\$11.1 billion, are far below the US\$40.8 billion of what is needed (Eberhard et al., 2011: 154).

Tackling existing utility inefficiencies, which include system losses, under-pricing, under-collection of revenue and over-staffing would make an additional \$8.24 billion available, but a funding gap would still remain (Eberhard et al., 2011:150).

IPPs have taken root in less than two dozen countries in Sub-Saharan Africa, or approximately half. Why have some countries, such as Uganda, been more successful than others in attracting IPPs? And which factors are important in enabling and sustaining private investment in power in developing countries? Building on pioneering work related to the obsolescing bargain and its application to private investment in electric power across the developing world (Woodhouse, 2006), Eberhard and Gratwick (2011) devised a set of country and project specific success factors that could partially predict the success of IPPs. At the country-level, the following factors were identified: stable country context; clear policy framework; transparent, consistent and fair regulation; coherent power sector planning; and competitive bidding practices (with further parameters detailed under each one). At the project level, these factors were generally as follows: favourable equity partners; favourable debt partners; creditworthy off-taker; secure and adequate revenue stream; credit enhancements and other risk management and mitigation measures; positive technical performance; and strategic management and relationship building. They also observed how an actual or perceived balance between investment and development outcomes contributed to project success and sustainability, across the sample of SSA countries with IPPs. This framework was based on a series of case studies and in depth country observations, dating to 2006, and further refined and revised through 2013 (Eberhard & Gratwick, 2005; Malgas, 2008; Gratwick, Ghanadan, and Eberhard 2006; Eberhard & Gratwick, 2012; Kapika & Eberhard, 2013). At the country-level, the investment climate, sector policies and reform, and regulatory certainty are all relevant – but recent research indicates

that more prosaic issues such as least-cost power planning linked to timely initiation of competitive procurement for new power are perhaps more significant (Eberhard et al., 2016). At the project level, traditional project finance concerns remain important – for example, equity and debt structuring, secure revenue flows, robust power purchase agreements with appropriate risk allocations, credit worthy off-takers or credit enhancement, guarantees and other security and risk mitigation mechanisms. In this paper, as we examine the record of IPPs in Uganda, we give particular attention to the relevance of planning and procurement issues in securing and sustaining private investment, including Chinese-funded projects. Closing Africa's power infrastructure funding gap inevitably requires undertaking reforms to reduce or eliminate system inefficiencies. This will help existing resources to go farther and create a more attractive investment climate for external and private finance, which still has the potential to grow. With the original drivers for market reform still present, private sector involvement appears inevitable in the future.

Finally, it is important to note that IPPs are not uniform. Although the typical IPP structure is understood as a privately sponsored project with nonrecourse or limited recourse project financing, IPPs in sub-Saharan Africa do not always follow this model. Instead, governments typically hold some portion of equity and/or debt or provide guarantees, bringing IPPs closer to a model of a public-private partnership (PPP) than that of the more traditionally conceived IPP. For the purposes of this study, IPPs were defined as power projects that are, in the main, privately developed, constructed, operated, and owned; have a portion (generally up to 1/3) of private finance; and have long-term power purchase agreements (PPAs) with a utility or another off-taker.

The history and structure of the Uganda electricity sector

The structure of Uganda's electricity sector reflects the ambitious reforms that began in the late 1990s (see Fig. 1 and Box 1). It is noteworthy that, although Uganda's electricity sector has been fully unbundled, all generation, transmission, and distribution assets are still state-owned, and the operation and maintenance of these facilities have been concessioned to private companies (Figs. 2 and 3).

Power sector reforms and their outcomes

Before the reform process began in 1999, the mismanagement and underperformance of Uganda's Electricity Board had created a sector that was underfinanced, worn-out, and delivering poor service (Wesonga, 2017). The core reform and restructuring process initiated by the 1999 Electricity Act lasted six years (MEMD, 2014). In 2002, the National Energy Policy was published to support the Act and facilitate private investment (MEMD, 2002). The policy calls for loans on concessionary terms, government guarantees and “smart subsidies” (grants) for power-sector investments.

The reform process was supported by a credit from the International Development Association (IDA) of the World Bank. The credit was extended under a program that promoted divestiture and restructuring of state-owned enterprises (SOEs), greater private sector participation, and strengthening of regulatory frameworks. A supplementary IDA contingent credit of \$5.5 million was made to support a liquidity facility for Umeme. The World Bank Group's (WBG's) Multilateral Investment Guarantee Agency (MIGA) extended insurance coverage for up to \$45 million in equity and shareholder loans to cover transfer restrictions, expropriation, war and civil disturbances, and breach of contract (Eberhard et al., 2016: 230–1).

By 2005, the Uganda Electricity Board had been unbundled into separate companies—UEGCL, UETCL, and UEDCL – and legislation provided for the expansion of each of these sub-sectors via varying degrees of private sector participation. International competitive tenders for the operation and maintenance of generation plants, and for the leasing

⁵ ODA has fluctuated between US\$1.57 and US\$1.78 billion over this period (World Bank, 2017b). However, the biggest change has been in the non-ODA borrowing. “Official development finance beyond ODA accounted for just 6.3% of total development finance to Uganda between 2002 and 2013, amounting to \$1.4bn. Since 2013, there has been a step change. In 2014–15 the Ugandan Parliament approved \$2bn of non-ODA loans, primarily from China. These made up 67% of total new external financing commitments for the year, including grants. Non-ODA loans are expected to constitute 70% of new government borrowings to 2025/26, amounting to \$7.4bn in value. Borrowing from China Exim Bank is expected to account for almost 80% of non-ODA loans to 2025,” (Davies et al., 2016:3).

⁶ The burden varies greatly by country, from 0.3% of GDP in Equatorial Guinea to 35.4% in Zimbabwe (Eberhard et al., 2011: 149).

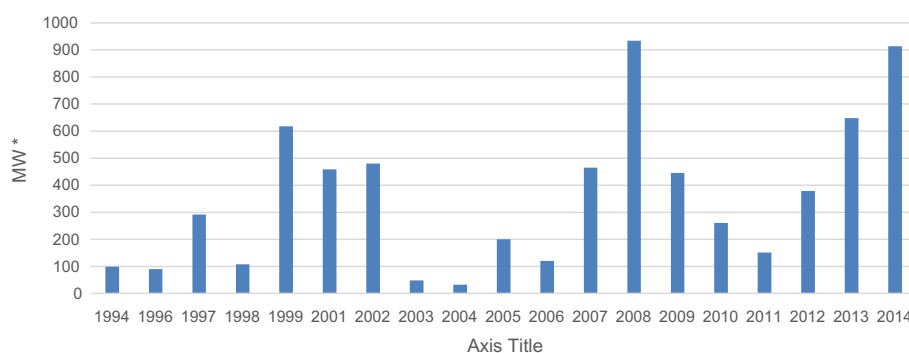


Fig. 1. Investment in independent power projects across SSA, 1994–2014 (excluding RSA). Note: MW based on financial close date, not on MW installed, e.g. in 1994 financial close was reached for the 99 MW CIPREL IPP in the Ivory Coast. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.) Source: compiled by authors.

Box 1

Uganda's main power sector institutions.

Ministry of Energy and Mineral Development (MEMD)

The MEMD is the focal point for energy policy matters within the Ugandan government. For public or emergency power generation projects, the MEMD continues to act as a procurement entity, either in its own right or through the sector's parastatals. Procurement for the ongoing Karuma and Isimba hydropower projects were directly handled by the MEMD (UEGCL was, however, given administrative power over the projects in 2016).

Electricity Regulatory Authority (ERA)

ERA's main responsibility is setting cost-reflective electricity tariffs. ERA also defines and monitors technical standards within the sector and enforces adherence to the National Grid Code. It issues and monitors the licenses required to generate, transmit, and distribute power. ERA also sets and reviews feed-in tariffs for renewable energy (REFIT) generation projects between 1 and 20 MW. In its capacity as a tendering authority under the Electricity Act, ERA conducted Uganda's first competitive tender for 20 MW of grid-connected solar concessions in 2014.

Uganda Electricity Generation Company Ltd. (UEGCL)

UEGCL is the holding company for state-owned generation assets. Its main roles are: to oversee the performance of ESKOM Uganda (see below) and of the thermal plant at Namanve (50 MW); to negotiate and administer engineering, procurement, and construction contracts for government/public projects as well as all Build Own Operate Transfer (BOOT) projects, namely IPPs; and to supervise the operations and maintenance of mid-tier public projects. UEGCL presently oversees Karuma and Isimba as well. UEGCL is also currently at the forefront of planning and procuring small PPP between 1 and 5 MW.

Eskom Uganda Ltd.

Eskom Uganda is a subsidiary of South Africa's utility giant Eskom Holdings. In 2003, Eskom Uganda was awarded a 20-year concession for the operation and maintenance of UEGCL's hydropower plants (Nalubaale, 180 MW, and Kiira, 200 MW).

Uganda Electricity Transmission Company Ltd. (UETCL)

The State-owned UETCL owns, operates, plans and procures Uganda's medium- and high-voltage transmission infrastructure (>33 kV). It also functions as the system operator, bulk single buyer (making it signatory to all PPAs), and dispatcher for almost all the electricity generated in Uganda (this excludes electricity generated within separate grids as the Electricity Act allows generators to sell directly to small energy co-operatives).

Uganda Electricity Distribution Company Ltd. (UEDCL)

UEDCL is Uganda's holding company for state-owned distribution assets, and it both administers and supervises the private distribution concession agreement (presently held by UMEME). UEDCL also operates a small number of mini-grids.

UMEME Ltd.

In 2005, UMEME Ltd. became the major privately owned electricity distributor in Uganda, after winning a 20-year concession to operate UEDCL's main distribution network. UMEME buys electricity at a bulk tariff from UETCL and, as of end-2016, they were selling it on to 950,814 customers. Businesses and government departments account for about 70% of the utility's annual revenue.

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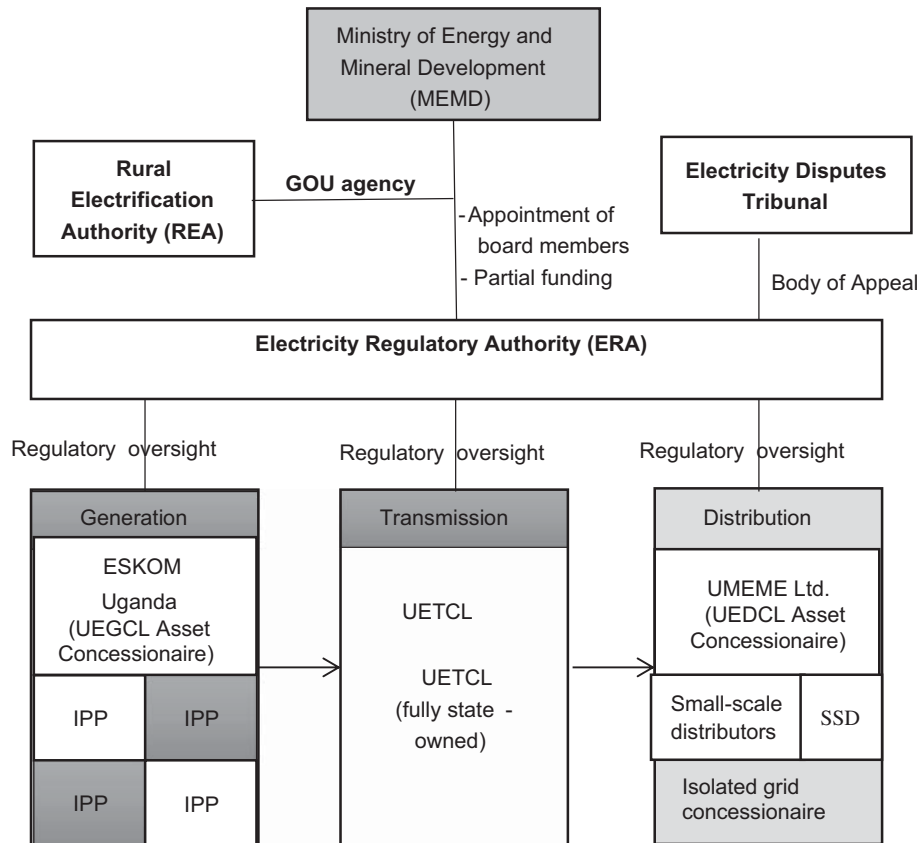


Fig. 2. The structure of Uganda's power sector.
Source: generated by authors.

of distribution assets, led to the awarding of concessions to ESKOM Uganda in 2003 and UMEME in 2005 (World Bank, 2009). Through these concessions, the government increased the funds available for rehabilitation, and incentivized good performance in accordance with private sector benchmarks. UETCL remained publicly operated but was removed from immediate government influence by being given an independent board and a corporate management structure. Despite plans to privatise UETCL, the government has so far refrained from this, due to political and policy reasons, and UETCL remains the sole off-taker of all electricity entering the main grid.

In 2006, seven years after the start of the restructuring process, its outcomes looked uncertain. The supply deficit had grown to between 90 and 210 MW (USAID, 2013), or approximately 50% considering the actual capacity was less than 500 MW at the time; load shedding was extensive and economic growth was being constrained. It had been hoped that the 250 MW Bujagali hydropower plant would be operational by this time. However, in 2004, corruption allegations led to the collapse of the contracted consortium, led by the US-based AES Corporation, and the project halted. In 2005, the procurement process was relaunched under the supervision of the World Bank and the European Investment Bank (EIB), and the plant was eventually completed in 2012.

In the meantime, the electricity supply shortage had to be mitigated by emergency thermal power, first supplied by Aggreko. At peak demand in 2011 and 2012, Uganda was generating up to 200 MW using heavy fuel oil and diesel (ERA, 2016a, 2016b, 2016c). The dispatch of these facilities drained UETCL's capital savings and affected its liquidity. The fact that the Ugandan shilling fell 25% against the US dollar in 2011, due to high energy prices, inflation and depreciation, also did not help. This, combined with the depletion of the World Bank's partial financing of thermal-based power production costs, led to a severe funding

shortfall in the power sector.⁷ Electricity tariffs were increased twice in 2006—by 41 and 35 respectively (Dhalla, 2011). However, despite severe and unsustainable drains on its resources, the government permitted no further tariff increases until 2012. In that year, end-user tariffs were increased by approximately 46%.⁸ This, together with power produced by Bujagali HPP after October 2012 (at approximately 11 US¢/kWh) gradually reduced the pressure on UETCL's balance sheet).

In an attempt to stimulate private investment in small-scale renewable energy projects, ERA adopted a Renewable Energy Feed-in Tariff (REFIT) scheme in 2012. Yet, it quickly became clear that problems remained, linked mainly to constraints on introducing cost-reflective tariffs for new projects. Financing and project development costs were high. Vital small-scale power projects proved unviable, even as the demand for power kept growing.

With another period of costly thermal-based power and depleting subsidies looming, the government responded in two ways. First, the procurement of roughly 800 MW of large hydro capacity was fast-tracked via financing and construction deals with China's Exim Bank and Chinese contractors (Foster, Butterfield, Chen, & Pushak, 2009). Second, for the small-scale projects, ERA and the German development bank, KfW, developed and implemented the Global Energy Transfer

⁷ In 2005, the government had to compensate UMEME for not being able to supply the amount of energy specified in the concession agreement, an additional drain on the national budget. In addition, until the 2011/2012 financial year, the government of Uganda paid a cumulative total of \$623 million in subsidies to UETCL. At their peak, the subsidies amounted to roughly 7% of Uganda's national annual budget. In the 2010/2011 financial year alone, the government paid more than US\$170 million in a direct subsidy to UETCL (SE4ALL, 2012).

⁸ Dhalla (2011) notes that while the weighted-average retail tariff in 2011 was \$0.126/kWh, a fully cost-reflective tariff would have been about twice that: \$0.251/kWh.

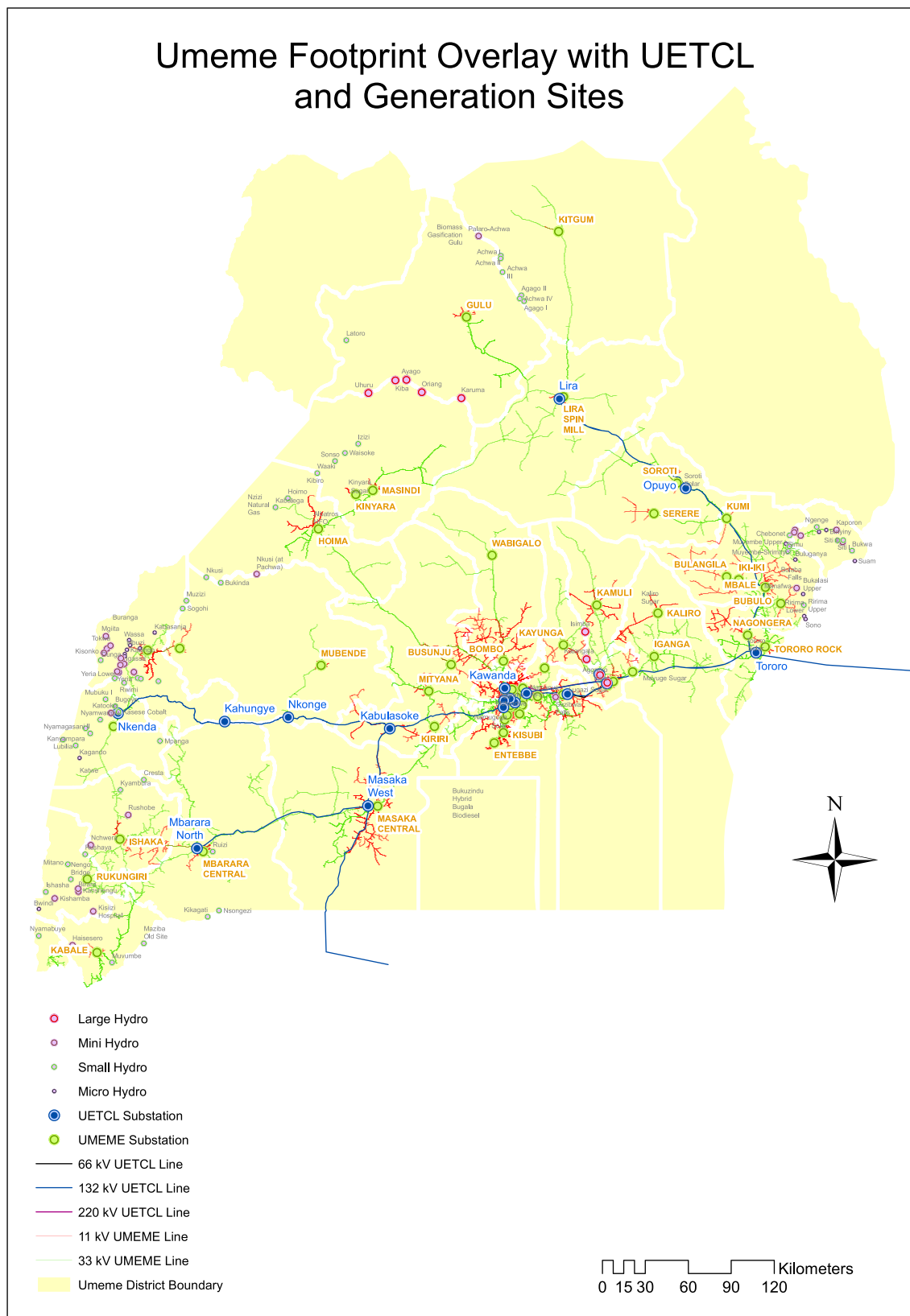
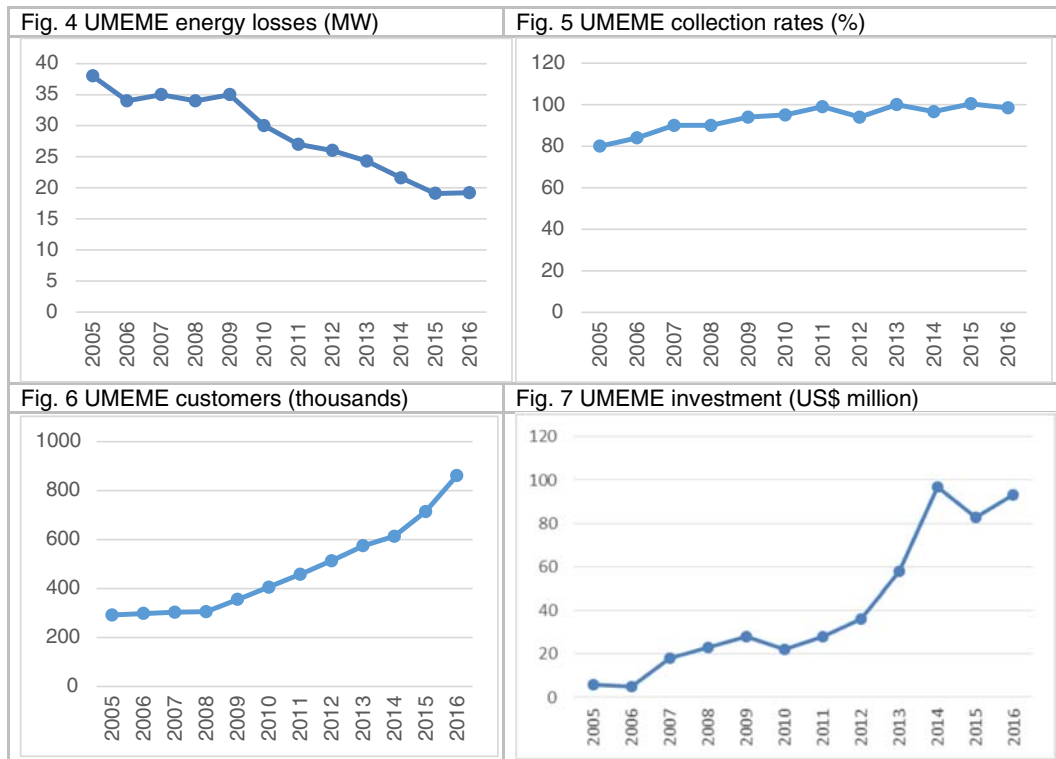


Fig. 3. Ugandan, electricity network, 2017. Note: the above map also contains prospective sites
Source: UMEME per comm, 2017.

Feed-in Tariff (GET FiT) program. Through this, up to 15 IPPs using various renewable energy sources and totaling 150 MW of generation capacity, were expected to become operational by 2018.

In 2016, Uganda's energy sector seemed to have moved beyond the most demanding phase of its transition and be sufficiently prepared for future challenges. As one indicator of sector health, UMEME's



Figs. 4–7. UMEME performance indicators. Fig. 4 UMEME energy losses (MW). Fig. 5 UMEME collection rates (%). Fig. 6 UMEME customers (thousands). Fig. 7 UMEME investment (US \$ million).

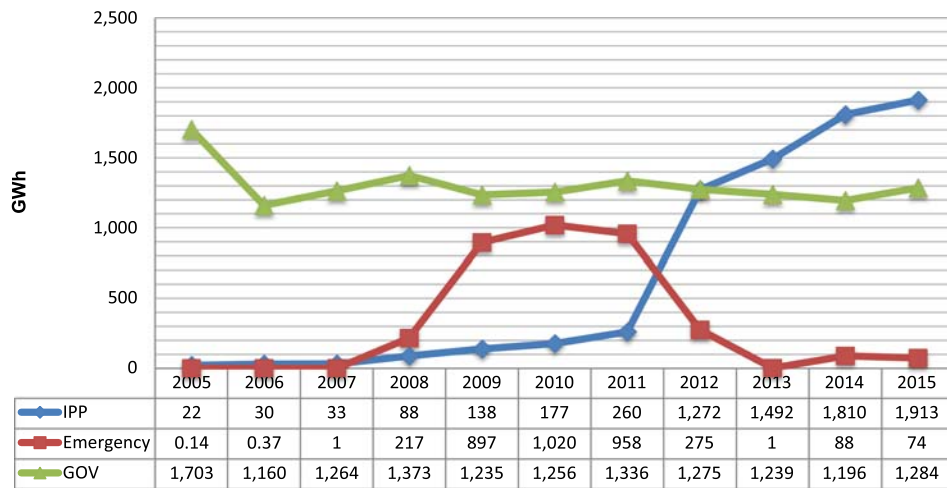


Fig. 8. Sources of electricity sold to UETCL, 2005–2015. Note: “Emergency” indicates thermal plant capacity that is used mostly for back-up power. It does not refer to short-term rental plants. The steep drop from 2012 to 2013 reflects the entry of the Bujagali hydroelectric plant into commercial service. Source: ERA Electricity Generation Statistics (2005–2016).

performance has steadily improved (demonstrated in Figs. 4–7, below). In 2012, UMEME listed its shares on the Uganda Securities Exchange through an initial public offering. More than 6,000 Ugandans bought stocks, as did African institutional investors, foreign equity funds, and venture capital funds. Funds raised from the stock offering were used to reduce the company's interest-bearing debt and enable UMEME to secure additional commercial debt over the next few years to help finance its expansion strategy.⁹

⁹ “An additional USD 60 m was drawn from the USD 170 m term facility with Standard Chartered Bank, Stanbic Bank Uganda Limited and International Finance Corporation (IFC), to fund the Company's capital investment plan for the year,” (UMEME, 2014: 13).

In 2015–6, large hydropower projects accounted for 70% of Uganda's power capacity (of approximately 850 MW), followed by thermal plants at 15%. Bagasse and small hydropower plants supplied equal shares of the remaining 14%. At that point, electricity production favoured IPPs (1,912 GWh) versus public plants (1,284 GWh), with a small share of thermal emergency capacity (see Fig. 8). (All conventional thermal capacity in Uganda—the Namanve and Tororo plants—is currently operated as emergency or standby capacity.) IPP production increased dramatically with the commissioning of the Bujagali HPP in 2012, which reduced the need for emergency power generation.

Table 1 reveals the expected evolution in the ownership and funding of Uganda's generation assets up to 2020. The commencement of

Table 1
Funding and ownership of generation capacity in Uganda in 2016 and 2020.

Ownership/funding	Installed capacity, 2016	Expected capacity, 2020
Government/state-owned utility	42%	24%
Government-owned (Chinese-built and funded)		45%
Emergency	15%	6%
IPP	42%	25%

operations at Bujagali resulted in a roughly even share of power generation between the public utility and IPPs. With the completion of Chinese-funded hydropower investments, by 2020, the share of public projects is expected to grow to roughly 70% of total installed capacity.

Power sector planning

Uganda's 1999 Electricity Act, the 2002 Energy Policy and the 2007 Renewable Energy Policy all affirm that the government (through the MEMD, UEGCL and ERA) holds primary responsibility for the expansion of generation capacity. MEMD's Energy Resources Department is responsible for forecasting national demand and supply and is also the main forum for the development of political consensus and decision-making on policies governing the electricity sector. While the legislation foresees no *direct* role for the government in developing or procuring new generation capacity, the MEMD has taken the lead in several procurements since 2005. This has increasingly resulted in concern, and, as will be shown in Section 6, has created some institutional problems for the sector as well.

Against this backdrop, ERA has exercised its role as the facilitator of private sector development to set up market mechanisms and competition for long-term contracts. Its influence has expanded in recent years, as evidenced by the 2012 end-user tariff hike, which ERA advocated to encourage further investment in the sector. Furthermore, the introduction of a quarterly automatic adjustment mechanism for end-user tariffs, which was promoted by ERA for years and which effectively floats electricity prices on the basis of macroeconomic parameters (beyond political control), could not have occurred without ERA's strong standing in the sector. As the sector's data collection hub, ERA has unrivaled insight into the market's subsectors and their dynamics. Armed with this knowledge, ERA has become the *de facto* policy advisor to all stakeholders (including the MEMD) in matters involving data and strategy. Beyond these advisory responsibilities, ERA implements sector policy in two important ways. First, it drives the development of the least-cost generation path (the recent Least Cost Development Plan was presented by ERA in 2015) even though this path has to be officially adopted by MEMD. Second, it affects sector planning by shaping the future energy mix by determining and enforcing capacity targets and limits, revising and implementing RE incentive frameworks (REFiT, competitive auctions) and by licensing generation projects based on marginal cost.

IPP procurement and the role of the government

The Electricity Act clearly stipulates that IPPs be developed (through unsolicited bids or competitive bidding for concessions) under ERA's direction, and that public Engineering, Procurement and Construction (EPC) projects will occur through parastatals such as UEGCL and UETCL. As per Sections 30 and 33 of the Electricity Act, ERA can either receive unsolicited bids or implement competitive bidding for concessions for IPP projects across all generation types. For unsolicited bids, ERA is the lead entity, and guides and monitors the planning and implementation of projects. ERA initiates procurements, conducts the necessary due diligence for the awarding of permits and licenses, and monitors the performance of all IPPs. For competitive tenders, such as

Table 2
Overview of available tax incentives for power generation investments in Uganda
Source: Compiled by authors, based on various primary and secondary source data.

Type	Details
Initial capital allowances	Initial allowance on plant and machinery of 50–75% ^a Start-up cost spread over 4 years (25% per year) Initial allowance of 20% on hotel, hospitals, and industrial buildings Deductible annual allowances (depreciable assets) of 20–40%
VAT	Exemptions for hydro (full/public and IPP ^b) Partial exemptions for solar (sole-purpose electromechanical equipment only)
Import duty/tax	Duty- and tax-free import of plant and machinery ^c Rebate of fuel duties Stamp duty exemption Exemptions from withholding on plant and machinery, scholastic materials, human and animal drugs, and raw materials Ten-year tax holiday
Repatriation of profits	No limits and no tax on repatriation of profits or dividends

^a This allowance was revoked in 2014. Now plant and equipment must qualify under the Income Tax Act for accelerated depreciation, with a maximum annual deductible of 20%.

^b Since 2014, costs incurred during the feasibility stage are no longer exempt from the value-added tax.

^c Only for certain electromechanical equipment

the solar generation implemented under the GET FiT program, ERA formally acts as the project executing agency.

Uganda made a relatively slow transition to the procurement of private power, first integrating excess capacity from several captive plants involved in mining operations. The first procurements of new capacity, following the 1999 reforms, were the Bujagali hydropower plant and various thermal power projects that were implemented between 2004 and 2010. During these years, the procedures laid out in the Electricity Act were largely ignored. Emergency thermal power plants were procured via a variety of players and contractual arrangements including (generally non-transparent) direct awards and (often non-transparent) competitive tenders.¹⁰ The various competitive tenders finally leading to the implementation of the Bujagali hydropower plant conducted between 2005 and 2008 were more transparent than previous private power deals cited above—carried out based on tender guidelines provided by the European Investment Bank and with significant managerial support and involvement by multilateral and regional development banks, including the World Bank (via IFC, IDA and MIGA), and the African Development Bank.

It is noteworthy that the Electricity Act specifies no direct role for the government (*viz.* MEMD) in developing or procuring new generation capacity. However, it was the MEMD that took the lead in awarding the two large-scale hydro projects—Karuma (600 MW) and Isimba (183 MW). Both Karuma and Isimba have since been transferred to UEGCL to avert further conflicts regarding project oversight between MEMD and UEGCL, (thus technically bringing the projects into compliance with the Electricity Act). But regulatory review and

¹⁰ The MEMD, through the UETCL, awarded the first contract to Aggreko in 2005 after competitive bidding. The second award to Aggreko for the Kiira project in 2005/06 was effectively the result of a direct negotiation process, one accompanied by allegations of secretiveness and mismanagement. With a total final price of \$300 million (installation + capacity payments), the Kiira 50 MW plant was four times as expensive as Aggreko's plant in Rwanda at \$74 million and twice as expensive as the \$160 million for the Lugogo 50 MW. Furthermore, stakeholders and observers characterized the procurement process "as rushed and shrouded in secrecy," which led the World Bank to pull out and instead provide IDA support for the \$206 million Mutundwe plant, which was commissioned in 2008 (<http://www.theeastafrican.co.ke/news/EAC-foots-huge-energy-bill-as-thermal-plants-have-a-field-day/-/2558/1226360/-/7dtem/-/index.html>, accessed February 1, 2015). The 2008 procurement of the Mutundwe project was partially supported by the IDA and implemented by the UETCL on behalf of the MEMD and World Bank. The contract for the Namanve plant, awarded to Jacobsen of Norway in 2008, was the result of a competitive procurement process under the Electricity Act, but this time implemented by ERA. The last thermal project followed a classic IPP model: Electro-Maxx's Tororo project resulted from an unsolicited bid process under Section 32 of the Electricity Act (1999).

Table 3

Feed-in tariff for hydro, bagasse and wind (2016–2018).

Source: Uganda renewable energy feed-in tariff guidelines (ERA (Electricity Regulatory Authority), 2016a, 2016b, 2016c)

Renewable energy technology	Tariff (US\$/kWh)	O&M (%) ^a	Cumulative capacity limits (MW)			Payment period (years)
			2016	2017	2018	
Hydro (10 > ≤20 MW)	0.094	10.96	30	60	80	20
Hydro (5 > ≤10 MW)	Linear tariff ^b	10.49	20	40	50	20
Hydro (500 kW > ≤5 MW)	0.107	10.49	10	20	30	20
Bagasse	0.088	29.78	30	50	60	20
Wind	0.122	10.71	25	50	75	20

^a The REFIT scheme also allows for an inflation indexation of the O&M share on an annual basis.^b Linear tariff, based on the actual installed capacity, was developed for mid-range hydro projects (from 10.7 US\$/kWh for 5 MW to 9.4 US\$/kWh for 10 MW).

oversight of these projects as required by the Electricity Act remains limited.

UEGCL regularly procures generation capacity for the government, often in cooperation with development partners, international financial institutions and private sector investors and developers in mixed equity public–private partnerships (PPP). Current examples of such procurements are the 46 MW Muzizi hydropower plant (for which KfW and other DevCO partners provided concessional loans and grants) and the 4.3 MW Nyagak III small hydropower station. For the latter, UEGCL entered into its first formal PPP arrangement with a local developer. However, the project has not yet been successful in raising the requisite capital, despite grant support provided by development partners.

Incentive frameworks

During the supply crisis of the late 2000s, the government introduced an array of incentives to encourage power sector investments (see Table 2). Among these were measures targeting renewable and thermal-based power investments, such as feed-in tariffs, take-or-pay arrangements (namely that the utility is bound to pay the contracted amount regardless of whether it actually takes the amount) and capacity deals (payment for capacity availed plus energy charge which guarantees a return so the seller does worry about market/dispatch risk). Additional supporting measures, such as sovereign guarantees and tax exemptions for general investments were introduced.

Fiscal incentives

REFIT incentives. For hydro, bagasse and wind projects with a capacity of between 1 MW and 20 MW, ERA established a feed-in tariff scheme that offers predefined, technology-specific off-take prices per kWh over the 20-year lifetime of the PPA, the terms of which are as shown in Table 3. This revision of the REFIT – introducing fully cost-reflective tariffs – was completed in 2016.

Despite the recent shift on the African continent and globally towards competitive auctions as a preferred tool for procurement of new RE-based generation capacity, ERA has decided to maintain the REFIT, only curtailing eligible technologies. ERA's continuous preference for REFIT is likely guided by the belief that technologies predominantly available in the country (small hydro and bagasse) are not ideally suited for competitive tendering due to e.g. their relatively small sizes (limiting investor interest) and location-specific characteristics. Feed-in tariffs for

other renewable technologies, namely biomass, biogas, landfill gas and geothermal were thus not set, allowing for bilateral negotiation or competitive tendering.

For biomass, biogas, landfill gas and geothermal, ERA assesses each project on its individual merit. Some parameters are, however, provided below (see Table 4) with regard to maximum return on equity and tariff ceiling. The rationale for exempting these technologies from REFIT is that they have not been tested on the national grid yet.

For solar PV, ERA fully abolished the initially available REFIT in 2012 and introduced competitive tendering under the Get FiT program.

GET FiT Uganda

After the 2012 REFIT failed to attract renewable energy investments due to absence of cost-reflective tariffs and bankable transaction documentation, the German Development Bank (KfW) together with ERA launched the GET FiT program. The aim of this program was to encourage new investments into small-scale RE projects that could be fast-tracked and mitigate the expected supply gap after 2016. The primary GET FiT mechanism is a grant-based premium payment paid as a top-up on the initial REFIT and designed to close the gap between the FiT with the levelized cost of electricity for eligible technologies, namely: small hydropower, biomass, bagasse and solar. The per kWh-based subsidy is calculated over the 20-year lifetime of the PPA, but is designed as a performance-based payment to the developer over the first five years of operation, thus enhancing the debt service profile.

Also part of the GET FiT program was the development of a full set of legal documents – including standardized technology-specific and investor-approved PPAs, implementation agreements, and direct agreements securing lender takeover rights. Furthermore, GET FiT assisted the government in streamlining other essential sectorial and regulatory procedures related to project implementation, such as permits and licensing, as well the tax and custom exemptions provided to IPPs. The GET FiT program also funded a secretariat, supported by an implementation consultant, who ran the tenders and assessed bids with ultimate approvals made by an independent investment committee.

For solar PV, GET FiT supported ERA in the design and implementation of the first competitive RE auction for IPP-promoted projects in East Africa (GET FiT Solar Facility). The targeted total installed capacity was 20 MW with a maximum project size of 10 MW per project. In absence of a REFIT for solar PV, GET FiT covered the gap between lowest tariff prices offered by developers in the auction and a previously agreed “base” tariff which would be paid by UETCL.

By 2Q2017, five tenders have been run for small hydropower, biomass, bagasse and solar PV projects. The total funding capacity availed by development partners (USD 90 million) is completely committed to 17 IPPs with a cumulated capacity of 157 MW. Eleven of the 17 projects have reached financial close. Four projects have been commissioned and an additional eight are being constructed after having reached financial close. It is expected that the balance – including one transboundary project with Tanzania – will be operational by 2018.

Table 4

Parameters for biomass, biogas, landfill gas and geothermal (2016–2018).

Source: Uganda renewable energy feed-in tariff guidelines, (ERA (Electricity Regulatory Authority), 2016a, 2016b, 2016c)

Renewable energy technology	Max return on equity %	Tariff ceiling (US\$/kWh)
Biogas	18	11.5
Landfill gas	18	8.9
Waste to energy/biomass	18	11.3

The procurement under GET FiT for REFiT technologies was based on quality rather than the price of projects (see Table 5). Bidders had to meet minimum qualitative benchmarks and were ranked in each tender according to their scoring in project-quality focussed criteria. The solar PV tender had a pricing element weighted with 70% of the total score.

As the first tender of its kind in East Africa and one of the pioneer auctions in Sub-Saharan Africa, the GET FiT Solar Facility generated overwhelming interest despite the small project size of 10 MW each. Twenty-four expressions of interest were submitted, out of which nine were shortlisted and invited to submit a full bid. The two solar projects awarded under the GET FiT scheme came in at an average cost of USc 16.4/kWh. This result is less impressive than has been achieved elsewhere (for example South Africa's renewable auctions led to prices as low as USc 6.4 kWh in the 4th round of bidding (Eberhard et al., 2016:179)), and can be attributed to the small market size, low level of competition, and Uganda's relatively high risk environment due to poverty, insecurity along borders, high corruption and inadequate infrastructure (Meyer, Tenenbaum, & Hosier, 2015). Nonetheless, the technology is gaining ground, and is still significantly cheaper than any alternative fossil fuel-to-power option, which range from USc 24–27/kWh (Eberhard et al., 2016:245). The first project – Access Uganda Solar Ltd.'s Soroti project – was operational 23 months after the launch of the tender.

Important to note is that the REFiT-based tenders attracted international and regional developers (including the Danish Private Equity Fund, Frontier, Sri Lanka's Eco Power, South Asia Energy Management Systems, based in the U.S., noted in Appendix A) in increasing numbers over the various tender rounds. In total, more than 30 applications for projects were received over four tender rounds.

GET FiT was designed as a temporary facility to stimulate the small-scale renewable energy market. The most important policy outcome – the development of a track record for small-scale IPPs and providing confidence to investors – has undoubtedly been achieved. In terms of future competitive tenders for RE, however, no announcements have been made by the regulator and there are reasons to assume that the GET FiT Solar Facility will remain the only auction-based procurement for the time being.

Chinese-funded projects and procurement priorities

Project evolution

As of 2Q2017, the first hydropower projects in Uganda to be primarily financed and built by China continue to be in the construction phase. Financing for both the 183 MW Isimba and the 600 MW Karuma plants, including upfront commitment payments (namely interest payments and transaction advisory costs) by the Ugandan government, were approved by Ugandan parliament in 2015 (AidData, 2015a, 2015b), with the Ugandan government contributing 15% of the total financing. Chinese contractors, SinoHydro and CWE, immediately began construction, though neither project is currently on track for commissioning in 2017 and 2018, respectively, as originally (and optimistically) expected. The third major hydropower scheme to be funded and built by China, Ayago (also 600–800 MW), was scheduled for launch in 2018, though this has since been postponed to 2023 (due to lack of funding and limited demand for the anticipated power).

Critical to acknowledge in this context, is that, the current and estimated future peak demand is below the lower base case assumptions in Uganda's integral power sector policy document (PSIP) dating to 2011. The capacity installed by 2020 after commissioning of the Karuma project (around a total of 2000 MW $\pm$ 10%) will likely be more than twice the domestic peak demand by this date. As nearly all recently added new generation capacity is based on "availed capacity" or "take-or-pay" arrangements, Uganda will be forced to pay for much more electricity than it effectively is able to dispatch.

The bidding and award processes used for Karuma and Isimba have taken various turns over the past two decades. The government's plans for Karuma were revised many times before a decision was made in 2009/10 to implement it as a public project. After the government decided to increase Karuma's planned capacity to 600 MW and procure a new feasibility study, support from Western donors waned over concerns about the environmental impact of the project, which lies on the boundary of one of Uganda's most pristine national parks (Barigaba, 2011).

Once the new consultant submitted the revised studies for Karuma, the MEMD implemented a new procurement process based on a public EPC model with project financing (Eberhard et al., 2016; Kasozi, 2010). Following the provisions of the Public Procurement and Disposal of Public Assets (PPDA) Act, the MEMD established evaluation and contract committees, staffed solely by senior officials from its own ranks and related sectoral institutions. Although the Act offers quite comprehensive guidelines on conducting public procurement, the process became highly politicized, and soon ran aground after allegations of bribery and other legal violations (Atuhaire & Akella, 2013; Wesonga, 2012). As a result, in September 2012 the PPDA Authority called for a halt to the process. In March 2013, the government's Inspectorate General also opened an investigation. Its report identified various violations of the procurement regulations. However, even though Uganda's High Court confirmed many of the violations assessed by the PPDA Authority, the tender process continued after a short re-evaluation of the technical bids. By this point, however, the procurement process had lost all public credibility and had to be suspended.

Confronted with the impasse, Uganda's president, Yoweri Museveni used the 2013 Durban BRICS conference and a meeting with Chinese President Xi Jinping to award the Karuma plant to SinoHydro and the Isimba to CWE. In particular the award to SinoHydro came as a surprise as the company had never been a short-listed bidder for Karuma, and it remains unclear on which assessments the award was obtained. The Isimba award alienated Western donors and finance institutions, which had believed the project was still earmarked for a public-private partnership. Nevertheless, the Chinese Exim Bank committed the required debt financing for both projects. It seems that the prospect of concluding an attractive financing deal for both projects swayed the Ugandan President and his officials. One comment frequently made by

Table 5

Evaluation criteria for global energy transfer feed-in-tariffs.

Source: Compiled by authors, based on various primary and secondary source data.

Classic GET FiT (small hydro, biomass, bagasse)	GET FiT solar facility
Financial and economic performance	Economic performance
Minimum financial internal rate of return, debt-service cover ratio, sensitivity	Economic rate of return
Dynamic production cost, economic rate of return, contribution to energy balance and grid stability	Project maturity and location
Environmental and social performance	Environmental and social performance
Quality and compliance with IFC rules on environmental and social impact assessment and environmental and social action plan	
Quality and compliance with IFC rules on resettlement action plan and livelihood restoration framework	
Technical and organizational performance	Technical and organizational performance
	Quality of technical documentation
Feasibility of proposed site	Project implementation timeline/expected commissioning date
Quality of technical documentation	Price proposed per kilowatt-hour
Project implementation timeline	(70% of total score)
Maturity of project and financial package	
Risk analysis	

Note: GET FiT = global energy transfer feed-in tariff; IFC = International Finance Corporation; kWh = kilowatt-hour.

Ugandan sector stakeholders has been the considerably larger “ticket size” (financing packages) offered by Chinese, which exceed the maximum loan amounts available from international financial institutions. The tedious coordination efforts and transaction costs occasioned by the multitude of financiers involved in the Bujagali project apparently had left a lasting impression on Ugandan government officials. Nevertheless, appealing though the deal may have been, it was almost certainly not in compliance with Ugandan law, and with the PPDA Act in particular.

In early 2014, the final financing conditions for both projects were presented to the Ugandan parliament. MPs approved the US\$1.4 billion loan agreement for Karuma in March 2015 with the Chinese ExIm Bank. The deal specified that the government would provide advance equity of 15% (US \$253 million) for Karuma, which the contractors used to kick off preliminary works. These funds were obtained from the dedicated reserves that the government had accumulated since 2007, and earmarked for generation capacity development.

Loan repayments will be made through electricity payments under a PPA with UETCL and UMEME. The Chinese PPAs for Karuma and Isimba have uniquely involved UMEME, as a contractual signatory to safeguard the repayment mechanism and avoid potential risks of revenues being misallocated. While the contractual arrangements have been finalized, capacity payments have not been made public, though the tariff is allegedly to be US\$ 4–6/kWh. However, this estimate is based on best case assumptions regarding cost and time overruns. Furthermore, the numbers reflect neither the significant out-of-pocket investment the government shouldered for Karuma and Isimba nor the looming contingent liabilities through sovereign guarantees issued to back the loans (World Bank, 2016:51; Kayondo, 2016; World Bank, 2015).

Besides the significant financial burden, the main sectorial challenge resulting from these awards is the sidelining of regulatory oversight and review. Even though the projects have in the meantime been legally and administratively transferred to UEGCL and are thus now formally subject to ERA's regulatory oversight, the previous conclusion of offtake and financing agreements through the government prevented the regulator from exercising its regulatory mandate and obligations, in particular in regard to review of the costs, tariffs and compliance with technical and quality standards. MEMD's actions have thus been judged “a challenge for the integrity of sector structures”.¹¹

As noted, the relevant legislation in Uganda favours competitive bidding, and direct negotiation of contracts such as EPC and finance awards for utility-scale hydropower projects have no legal foundation. As a matter of fact, the handshake agreements for the Karuma and Isimba plants contravene applicable law. However, unlike the initial bid process related to the Karuma plant, the decisions have not been challenged by international developers or domestic stakeholders. The projects are going ahead, with contracts awarded to experienced contractors (e.g. Sinohydro and CWE) who initially gave realistic implementation timelines. Just six months lapsed between the awards being made and the first steps towards final engineering design studies. Six months later, after the provision of the substantial advance payment by the government, the projects were in the early stages of construction, and the government seems to have secured the nation's energy supply.

Shifting procurement policies

For the future, the Ugandan government seems intent on following a two-pronged generation procurement policy. For large-scale projects, international competitive bidding seems to have been abandoned in favor of direct awards to international—effectively Chinese—contractors. On the other end of the scale, targeted policies (promoted particularly by ERA) aim to encourage foreign investment in small to medium sized IPP projects involving all types of generation. Attempts

to facilitate small-scale renewable energy projects and the implementation of the GET FiT scheme are evidence of this. In general, the Ugandan government seems to see competitive bidding and IPPs as too costly and time-consuming for large-scale projects.¹² This perception has been shaped by comparisons of the cost and implementation timelines of the government-owned Nalubaale and Kiira plants with the privately sponsored Bujagali facility, but this comparison merits further analysis.

Whereas the Nalubaale and Kiira projects sell electricity to the utility at an estimated US\$ 1.2/kWh, Bujagali sells its electricity at about US\$ 11/kWh. For the Chinese-backed Karuma and Isimba projects still under construction, the government has publicly communicated an expected tariff range of US\$ 4–6/kWh. Development partners as well as the private sector have questioned these numbers and, indeed, a closer look at cost estimates and financing conditions seem to challenge the government's expectations. At US\$3.44 million per MW, the Bujagali hydropower plant (HPP) ranks among the most expensive projects of its scale in the world (IRENA, 2012). Karuma's likely costs are estimated at a more effective level of \$2.34 million per MW, but Isimba, at \$3 million per MW is not significantly cheaper than the Bujagali plant at a per unit level. Cost overruns on large hydropower projects are common, so it is possible that the margin of public over private projects in Uganda could decrease further, making the government's cost-based argument for pursuing direct awards baseless. In addition, the lack of transparency surrounding the award and its negotiation could further inflate the final total costs and “hide” the real costs outside of the tariff.

It can also be argued that the IPP approach is more beneficial for the national fiscus, at least in the short term: whereas, the government has had to make an advance payment of roughly US\$320 million (15% of estimated total project costs for 600 MW) for the Chinese-promoted project, its equity contribution to the Bujagali plant was only US\$20 million (2.3% of the total cost for 250 MW). Although Uganda had accumulated sufficient reserves to cover the US\$320 million, it does mean that significantly less capital is available for other much-needed strategic investments in the energy sector.

The government's other main argument in favor of a direct awards relates to their comparatively shorter implementation timelines. Public perception is that the full procurement cycle for Bujagali took over 12 years. In contrast, the implementation of the similarly sized Kiira plant in the early 2000s is perceived—albeit wrongly—as having proceeded without complications or delays. The Karuma and Isimba projects were projected to take just six years from award to expected commissioning. At face value, no transparent competitive bidding process, conforming with international legalities and formalities can compete with this, particularly if international financiers are involved. It can, however, be argued that procurement process itself was less responsible for the delays associated with Bujagali, but rather problems with the institutional arrangements responsible for its implementation, including the exclusion of experts from the procurement and decision-making process, were to blame. In particular, Ugandan government officials seem to overlook the fact that the first attempt to develop the Bujagali plant failed after corruption in the direct negotiation process was exposed. Similarly, the often praised Kiira plant has been running well below its actual installed capacity, which may mainly be attributed to an inadequate due diligence of water resource availability – a shortfall which an international investor-backed IPP procurement process would have likely avoided.

Assessing the outcomes of the reform process

Despite the recent shift towards publicly-initiated procurements, for the moment IPPs still thrive in Uganda. The recent confirmation of

¹² There are only two other examples of large scale hydro IPPs in SSA: Bujagali (250 MW) and Zambia's Itzhi (120 MW), however, there are been numerous OCGT, CCGT IPPs in excess of 100 and 200 MW.

¹¹ Pers. comm. with ERA, November 2014.

Uganda's first fully cost-reflective REFiT underscores the regulators' commitment to maintain a momentum for IPPs. Various new projects have been proposed and approved by ERA, including a 50 MW solar PV project, two 40 MW hydro projects as well as various new small hydropower, bagasse and biomass plants. The prospects for fossil fuel IPPs will likely improve once Uganda manages to tap into its considerable crude oil resources. International investors have shown interest in developing and financing natural gas-based projects.

In general, the Ugandan government has been successful in achieving its development goals for the power generation sector. In the late 2000s, the sector faced the challenge of attracting international investment to replace costly diesel-based generation, and to avert financial disaster for the state utilities. The commissioning of the Bujagali plant in 2012 eased pressures on the Ugandan government considerably—albeit only on the supply side; consumer tariffs remained problematic. Since then, the sector's strategy has consolidated, and more forward-looking policies were reflected in decision making. Support for small-scale renewable energy projects and the implementation of the GET FiT programme was based on a desire to mitigate the interim supply shortages. By early 2017, with close to 1,000 MW in development, generation capacity under development has multiplied within just five years.

Uganda has also managed to develop a mix of public projects financed by Chinese sources and privately financed small-scale IPP projects—a combination that is unique in the region. At a bird's eye view, Uganda's development and investment outcomes appear balanced, at present. The increase in investor interest in Uganda is tangible. [Bloomberg New Energy Finance ranked the country seventh in a 2016 global survey of investment climate in 58 emerging economies \(and second in Africa, after South Africa\).](#)

With regard to [Eberhard and Gratwick \(2012\)](#), Uganda's successes rely less on its overall investment climate and more on the clarity of the structures in its power sector structure. The fact that Uganda's power sector has been unbundled allows IPPs to contract directly with the transmission company, unconstrained by potential conflicts with state-owned generation facilities. Thus, there is openness to private sector participation and a clear policy framework. Furthermore, the privately concessioned distribution company has steadily reduced both technical and commercial losses and thereby improved its financial viability as well as helped the sector more generally. Uganda has one of the few financially sustainable electricity sectors in Sub-Saharan Africa, in which tariffs are cost reflective and do not require government subsidies ([Kojima & Trimble, 2016](#)), which also helps to attract private sector investment into generation. An independent regulator has also been integral to attracting IPPs and sustaining the sector. ERA has introduced a tariff adjustment which automatically adapts retail tariffs to a certain set of macroeconomic indexes, including the USD-Shilling exchange rate and the oil price. This ensures that returns of the utilities are stable. More importantly however, it removes necessary short-term retail tariff adaptations from the political sphere.

These round out what Eberhard and Gratwick termed 'country level' contributing elements of success with regard to Ugandan IPPs, at present. At the project level, the GET FiT programme has done much to support competitive tenders, though it is not anticipated that this initiative will be sustained. Prior to GET FiT, Uganda had generally attracted two or three bidders—less than are needed to ensure strong competition. However, as demonstrated by the GET FiT programme, multiple bid rounds enabled progressive improvement of documentation and contracts; they build investor confidence and a pipeline of bankable projects, which can more easily reach financial close and commissioning.¹³ DFI involvement in project debt has also been a deal maker at the project level, as have robust PPAs in the context of GET

FiT and experienced project partners that are committed to working in Africa and assuming the associated risk.

Outlook

There is, however, another side to Uganda's electricity reform and assessing IPP outcomes, which involves a different time horizon, and a good look at the future. For every accomplishment along the power sector reform path, a price was paid. For example, reaching cost recovery and stability was a painful experience for many stakeholders. The "shock therapy" with various steep tariff hikes between 2008 and 2012 had a lasting impact on consumer wallets. In the perception of many government officials as well, private sector involvement is to blame for these steep increases ([Khisa, 2017](#)). Thus, the operations of UMEME Ltd. are meticulously scrutinized by parliament and various demands have been made to the government to revoke the concession. Similarly and as previously described, private sector involvement in the generation sub-sector is viewed as too costly although only time will tell whether the currently implemented, Chinese-funded hydropower projects will be able to deliver electricity at cheaper rates as claimed by the government. There are reasons to believe that – actual costs considered – this will likely not be the case.

Although the country has met its short-term supply needs, the future presents a very different challenge, calling in to question the effectiveness of electricity planning in Uganda. In nearly every case that [Eberhard and Gratwick \(2011\)](#) have studied in Africa, planning has emerged as the single most important success factor for long-term viability of IPPs and the same is seen in Uganda. Uganda's effective demand for electricity has not developed as projected a decade ago. The reasons for this mismatch in supply and demand are many, as highlighted earlier, including the pending industrialization of the country, the absence of expected large-demand consumers from the oil industry as well as lagging progress in electrification and change in consumer demand patterns in off-peak hours. The relatively small size of the Ugandan power system already constrains its ability to procure reasonably priced IPP capacity, without the additional supply-demand mismatches.

Publicly, the government claims that Uganda will be able to export excess electricity to neighbouring countries in line with what is labelled a "battery of the region" policy. However, this assumption neglects the fact that neighbouring countries – perhaps with the exceptions of South Sudan and the Democratic Republic of Congo – have also been adding capacity.

Tanzania, Kenya and Rwanda also expect to become net exporters by the time Uganda's excess capacity will reach its peak. The final blow to the export policy could come from electricity fed into the East African Power Pool by Ethiopia, whose large hydropower-based capacity produces at a fraction of the price of Uganda's current and future generation assets. The government will thus need to plan more far-sighted to prevent the worst outcomes – and do so fast. Even a temporary excess supply of the expected scale has undoubtedly the potential to drive the offtaker, UETCL, to bankruptcy in a matter of months. The government will face the conundrum between providing direct subsidy support to the utility or allow it to default on payments and trigger the sovereign guarantees provided to all projects developed since 2012. Uganda might thus end where it started from: providing significant direct or indirect fiscal subsidies for an otherwise bankrupted sector. In a worst-case scenario, a "race" for payments from sovereign guarantees could occur, in which case small and medium-scale IPPs would likely be the weakest contenders.

Additional deemed energy payment obligations may arise from implementation delays for critical network expansions and rehabilitations. These are required to ensure the full evacuation of IPPs currently under development, namely that if IPPs, after commissioning, are prevented from delivering electricity to UETCL then they will still be compensated for what they could have delivered. Despite the "ticking clock" resulting

¹³ There has been considerably more activity outside of RSA since 2015. In that year, the Scaling Solar program in Zambia attracted 48 bids, with 11 qualifying firms and 7 tenders submitted. Namibia and Mauritius have also seen notable activity, as has Ghana.

from these deemed energy clauses in already executed PPAs, and significant funding availed by development partners, progress has been slow, mainly due to managerial and operational shortfalls by UETCL and REA, the implementing agencies. Ironically, both government and the offtaker now pass the blame to the private sector which required deemed energy obligations to ensure the bankability of their investments. As a consequence of this development, the initially embraced standardized transaction documentation developed under GET FiT are currently under review again. The likely outcome of this process is that the existing bankable risk allocation offered will be abolished. Due to the government's obvious policy shift towards publicly financed large scale projects, other incentive frameworks provided to the private sector may also come under scrutiny. The well-balanced policy package offered to developers and investors over the last few years might thus soon be shelved.

Similarly, permanent discussions around terminating or not extending concessions awarded to private sector actors for both UEGCL's

generation assets as well as UEDCL's main distribution grid give a clear indication of the general trajectory of broader government policy considerations. The Ugandan government is certainly not to be criticized, neither for its goal of reducing cost of electricity supply nor for its aim of building local technical capacities for operation of public projects. And understandably, there is some regret for “paying a premium” for being a Sub-Saharan early mover in private sector investment in the electricity sector.

However, ultimately shortfalls in far-sighted and informed policy planning and implementation phasing are currently jeopardizing the future financial and economic sustainability of Uganda's electricity sector. In particular, the award of an additional 800 MW through the Karuma and Isimba projects in 2012 came at a time when constrained demand growth was already perceivable and it was thus likely that excess supply would become inevitable. Private sector investors are just too appealing a scapegoat.

Appendix A. IPPs in Uganda

Financial closure year	Project name	Project status	Technology/fuel	Total Investment (US\$ million)	Capacity (MW)	Award method ^a	Number of bids	COD	Contract period	Contract type	Sponsors/developer
1999	Kasese cobalt, (Mubuku III)	Operational	Hydro, Small (<20 MW)	22.50	9.9	DN			20		Blue Earth Refineries (100% Uganda)
2003	Kakira cogeneration plant	Operational	Waste/bagasse	56.00	32	DN	NA	2013	20	BOO	Madhvani Group (100% Uganda)
2015	Kakira cogeneration plant expansion	Operational	Waste/bagasse	57.00	20	REFIT	NA	2015	20	BOT	Madhvani Group (100% Uganda)
2007	Bujagali hydro project	Operational	Hydro, large	860.00	250	ICB	3	2012	30	BOT	BEL Ltd. (Sithe Global Power (58% United States), Aga Khan Fund (31% Switzerland))
2008	Tronder/Bugoye hydroelectric power project (Mubuku II)	Operational	Hydro, Small (<20 MW)	65.70	13	DN	NA	2009	20	BOT	Tronder Power Limited (100% Norway)
2008	Mpanga hydropower project	Operational	Hydro, Small (<20 MW)	27.00	18	DN	NA	2011	20	BOT	South Asia Energy Management Systems (SAEMS) (100% United States)
2008	ECO Ishasha mini hydropower plant	Operational	Hydro, small (<20 MW)	14.00	6.5	DN	NA	2011	30	BOT	Eco power (100% Sri Lanka)
2008	Namanve power plant	Operational	MSD/HFO	74.00	50	ICB	3		6	BOT	Jacobsen Elektro (100% Norway)
2009	Tororo power station	Operational	MSD/HFO	41.50	16	DN	NA	2010	9	BOO	Electro-maxx (100% Uganda)
2012	Tororo power station expansion	Operational	MSD/HFO	41.50	34	DN	NA			BOO	Electro-maxx (100% Uganda)
2009	Buseruka/hydromax hydropower plant	Operational	Hydro, small (<20 MW)	27.00	9	DN	NA	2012	30	BOT	Hydromax limited (100% Uganda)
2009	Kinyara cogeneration plant	Operational	Waste/bagasse	29.00	7.5	DN	NA	2011	20	BOO	Kinyara sugar group (100% Uganda)
2013	Kilembe mines (Mubuku I)	Operational	Hydro, small (<20 MW)	16.20	5.4	DN			25		Tibet-Hima consortium (100% China)
2015	Siti I hydropower project	Operational	Hydro, small (<20 MW)	14.80	6.1	REFIT	NA	2017	20	BOT	Frontier energy (Danish Private Equity Fund)
2015	Rwimi hydropower project	Construction	Hydro, small (<20 MW)	20.80	5.4	REFIT	NA	Exp 2017	20	BOT	Eco power (100% Sri Lanka)
2015	Muvumbe hydropower project	Operational	Hydro, small (<20 MW)	14.10	6.5	REFIT	NA	2017	20	BOT	Vidullanka (100% Sri Lanka)
2016	SAEMS Nyamwamba SHPP	Construction	Hydro, small (<20 MW)	26.80	9.2	REFIT	NA	Exp. Q1 2018	20	BOT	South Asia Energy Management Systems (SAEMS) (100% United States)
2016	Siti II hydropower project	Construction	Hydro, small (<20 MW)	33.00	16	REFIT	NA	Exp. Q3 2018	20	BOT	Frontier energy (Danish Private Equity Fund)
2016	Lubilia hydropower project	Construction	Hydro, small (<20 MW)	18.70	5.4	REFIT	NA	Exp. Q4 2017	20	BOT	Frontier energy (Danish Private Equity Fund)
2016	Waki	Construction	Hydro, small (<20 MW)	18.10	4.8	REFIT	NA	Exp. Q2	20	BOT	Hydromax limited (100% Uganda)

(continued on next page)

Appendix A (continued)

Financial closure year	Project name	Project status	Technology/fuel	Total Investment (US\$ million)	Capacity (MW)	Award method ^a	Number of bids	COD	Contract period	Contract type	Sponsors/developer
2016	Soroti Solar	Operational	Solar, PV	19.00	10	REFIT	NA	2018	20	BOT	Access (UAE)/TSK (Spain)
2016	Tororo Solar	Operational	Solar, PV	20.00	10	REFIT	NA	2017	20	BOT	Access (UAE)/TSK (Spain)
2017	Sindila	Construction	Hydro, small (<20 MW)	17.00	5	REFIT	NA	Q3 2018	20	BOT	Lereko Metier sustainable capital (87%) and KMRI LLC USA; WK power and fieldstone Africa.
2017	Kyambura	Construction	Hydro, Small (<20 MW)	24.00	7.6	REFIT	NA	Q4 2018	20	BOT	
2017	Nkusi	Construction	Hydro, small (<20 MW)	23.00	9.6	REFIT	NA	Q1 2018	20	BOT	PA Technical Services

^a International competitive bidding (ICB) direct negotiation (DN).

Appendix B. Chinese-funded generation projects in Uganda

Financial closure year	Project name	Project status	Technology/typology	Total investment (US\$ million)	Capacity (MW)	Award method	COD	Developer/EPC	Chinese funding source bank
2015	Karuma hydropower project	Construction	Hydro, large	1653.00	600	DN	Exp. 2019	Sinohydro	China Eximbank
2015	Isimba hydropower project	Construction	Hydro, large	556.00	183	DN	Exp. 2018	CWE/Sinohydro	China Eximbank

References

- Adamantiades, A. G., Besant-Jones, J. E., & Hoskote, M. (1995). Power sector reform in developing countries and the role of the World Bank. *16th congress of the world energy council, Tokyo*. Washington, DC: Industry and Energy Department, World Bank.
- AidData (2015a). China EXIM Bank commits \$789.3 million USD loan for Karuma dam construction project in Uganda (tranche 1/2). Available via: <http://china.aiddata.org/projects/30919>, Accessed date: 1 June 2017.
- AidData (2015b). China EXIM Bank commits \$645.82 million USD loan for Karuma dam construction project in Uganda (tranche 2/2). Available via: <http://china.aiddata.org/projects/36211>, Accessed date: 1 June 2017.
- APEC Energy Working Group (1997). *Manual of best practice principles for independent power producers*. Canberra: APEC Secretariat.
- Atuhaire, A., & Akella, J. (2013). Questions over Karuma dam, the independent. October 4, 2013. Available via: <https://www.independent.co.ug/questions-karuma-dam/>, Accessed date: 16 October 2017.
- Bacon, R. (1995). Privatization and reform in the global electricity supply industry. *Annual Review of Energy and the Environment*, 20, 119–143.
- Bacon, R. (1999). *A scorecard for energy reform in developing countries*. Washington, DC: Finance Private Sector and Infrastructure Network, World Bank.
- Barigaba, J. (2011). Donors query Uganda on excess capacity in Karuma dam. *The East African* July 10 2011. Available via: <http://www.theeastafrican.co.ke/news/Donors-query-Uganda-on-excess-capacity-in-Karuma-dam/2558-1197970-61c5abz/index.html>, Accessed date: 11 October 2017.
- Besant-Jones, J. E. (2006). *Reforming power markets in developing countries: What have we learned?* Energy and mining sector board discussion paper no. 19. Washington, DC: World Bank.
- Bloomberg New Energy Finance (2016). *Climatescope 2016*. Available via: <http://global-climatescope.org/en/>, Accessed date: 4 May 2017.
- Castellano, A., Kendall, A., Nikomarov, M., & Swemmer, T. (2015). Brighter Africa: The growth potential of the sub-Saharan electricity sector. Available via: <http://www.mckinsey.com/industries/electric-power-and-natural-gas/our-insights/powering-africa>, Accessed date: 1 November 2017.
- China Africa Research Initiative (CARI) (2016). *Chinese loans in Africa. School of advanced international studies*. Washington D.C.: Johns Hopkins Interim document.
- Davies, F., Long, C., & Wabwire, M. (2016). *Age of choice: Uganda in the new development finance landscape*. London: Overseas Development Institute.
- Dhalla, G. (2011). Performance of the Ugandan power sector to 2017. *Report submitted to the Ministry of Energy and Minerals Development, Kampala*.
- Eberhard, A., & Gratwick, K. (2012). *Light inside: The experience of independent power projects in Nigeria*. Tunis: Infrastructure Consortium for Africa Working Paper.
- Eberhard, A., Gratwick, K., Morella, E., & Antmann, P. (2016). Case study 5: Power generation developments in Uganda. *Independent power projects in sub-Saharan Africa: Lessons from five key countries*. Washington, DC: World Bank.
- Eberhard, A., & Gratwick, K. N. (2011). IPPs in sub-Saharan Africa: determinants of success. *Energy Policy*, 39(Issue 9), 5541–5549.
- Eberhard, A., Rosnes, O., Shkaratan, M., & Vennemo, H. (2011). *Africa's power infrastructure: Investment, integration, efficiency*. Washington, DC: World Bank.
- Eberhard, A. A., & Gratwick, K. (2005). The experience of independent power producer investments in Kenya. *Journal of Energy in Southern Africa*, 16(4), 152–165.
- ERA (Electricity Regulatory Authority) (2016a). *Electricity generation statistics (2005–2013)*. Kampala: ERA.
- ERA (Electricity Regulatory Authority) (2016b). *Electricity generation statistics (2014–2016)*. Available via <http://www.era.or.ug>, Accessed date: 2 May 2017.
- ERA (Electricity Regulatory Authority) (2016c). *Uganda renewable energy feed-in tariff guidelines*. Kampala: ERA.
- ERA (Electricity Regulatory Authority) (2017). *The electricity supply industry of Uganda*. Kampala: ERA.
- ESMAP (Energy Sector Management Assistance Programme) (1999). *Rural electrification strategy study. Report 221/99*. Washington D.C.: World Bank Available via: <http://documents.worldbank.org/curated/en/521531468779364234/pdf/multi-page.pdf>, Accessed date: 11 October 2017.
- Foster, V., Butterfield, W., Chen, C., & Pushak, N. (2009). *Building bridges: China's growing role as infrastructure financier for sub-Saharan Africa*. Washington, DC: World Bank.
- Gratwick, K., Ghanadan, R., & Eberhard, A. (2006). "Generating Power and Controversy: Understanding Tanzania's Independent Power Projects.". *Journal of Energy in Southern Africa*, 17(4), Cape Town: Energy Research.
- Gratwick, K. N., & Eberhard, A. (2008). Demise of the standard model for power sector reform and the emergence of hybrid power markets. *Energy Policy*, 36, 3948–3960.
- IRENA (International Renewable Energy Agency) (2012). *"Hydropower." Renewable Energy Technologies: Cost Analysis Series 1 (3/5)*. IRENA: Bonn.
- Jhirad, D. (1990). Power sector innovation in developing countries: implementing multi-faceted solutions. *Annual Review of Energy*, 15, 365–398.
- Kapika, J., & Eberhard, A. (2013). Uganda: Brave reforms and new growth. *Power-sector reform and regulation in Africa: Lessons from Kenya, Tanzania, Uganda, Zambia, Namibia, and Ghana*. Cape Town: Human Sciences Research Council Press.
- Kasozzi, E. (2010). Karuma hydro power project raises capacity. *Daily monitor* December 21, 2010. Available via: <http://www.monitor.co.ug/Business/Prosper/688616-1075640-h099if/index.html>, Accessed date: 1 November 2017.
- Kayondo, S. (2016). Uganda should consider enacting a sovereign debt restructuring policy. *Daily monitor* 25. Available via: <http://www.monitor.co.ug/OpEd/Commentary/Uganda-enacting-sovereign-debt-debt-restructuring-policy/689364-3256866-oti47/index.html>, Accessed date: 16 October 2017.
- Kessides, I. N. (2004). *Reforming infrastructure*. Washington, D.C.: World Bank and Oxford University Press.
- Khisa, I. (2017). Can Uganda survive with highest electricity prices in EA? *The independent* January 31 2017. Available via: <https://www.independent.co.ug/can-uganda-survive-with-highest-electricity-prices-in-ea/2/>, Accessed date: 16 October 2017.
- Kojima, M., & Trimble, C. (2016). *Making power affordable for Africa and viable for its utilities*. Washington, DC: World Bank.

- Malgas, I. (2008). *Energy stalemate: Independent power projects and power sector reform in Ghana. MIR working paper*. Cape Town, University of Cape Town, Graduate School of Business.
- Malgas, I., & Eberhard, A. (2011). Hybrid power markets in Africa: Generation planning, procurement and contracting challenges. *Energy Policy*, 39(2011), 3191–3198.
- MEMD (Ministry of Energy and Mineral Development) (2002). *Energy policy for Uganda*. Available via <http://energyandminerals.go.ug/downloads/EnergyPolicy.pdf>, Accessed date: 2 May 2017.
- MEMD (Ministry of Energy and Mineral Development) (2014). *2014 statistical abstract*. Available via <http://www.energyandminerals.go.ug/downloads/2014StatisticalAbstract.pdf>, Accessed date: 1 November 2017.
- Meyer, R., Tenenbaum, B., & Hosier, R. (2015). *Promoting solar energy through auctions: The case of Uganda. Live wire knowledge note series, no. 2015/49*. Washington, DC: World Bank Group.
- Moore, E. A., & Smith, G. (1990). *Capital expenditures for electric power in the developing countries in the 1990s*. Washington DC: The World Bank.
- Okoboi, G., & Mawejje, J. (2016). Electricity peak demand in Uganda: insights and foresight. *Energy, Sustainability and Society*, 6, 29.
- Patterson, W. (1999). *Transforming electricity*. London: Royal Institute of International Affairs.
- SE4ALL (2012). *Uganda country profile*. Kampala Available via <http://www.se4all.org>, Accessed date: 1 June 2015.
- UMEME (2014). *Annual Report 2014*. Kampala: UMEME.
- USAID (U.S. Agency for International Development) (2013). *Africa infrastructure program. Uganda power sector assessment report*. Washington, DC: USAID.
- Victor, D., & Heller, T. C. (Eds.). (2007). *The political economy of power sector reform: The experiences of five major developing countries*. Cambridge: Cambridge University Press.
- Wesonga, N. (2012). 'Chinese firm in Karuma project scandal 'deletes' disputed information,' daily monitor 25. Available via: <http://www.monitor.co.ug/News/National/688334-1395050-9hlnfqz/index.html>, Accessed date: 16 October 2017.
- Wesonga, N. (2017) Special report — Impact of unbundling: power sector. Tracing growth of Uganda's power sector. Daily monitor. May 5, 2017. Available via: <http://web.monitor.co.ug/Supplement/2017/PowerSector20062017.pdf>, accessed on October 16, 2017.
- Williams, J. H., & Ghanadan, R. (2006). Electricity reform in developing and transition countries: a reappraisal. *Energy*, 31, 815–844.
- Wolak, F. A. (1998). *Market design and price behavior in restructured electricity markets: An international comparison*. (Conference on Electricity Industry Restructuring. Berkeley).
- Woodhouse, E. (2006). 'The obsolescing bargain redux? Foreign investment in the electric power sector in developing countries', N.Y.U. *Journal of International Law and Politics*, 38, 121–219.
- World Bank (1993). *The World Bank's role in the electric power sector: Policies for effective institutional, regulatory, and financial reform*. Washington, DC: World Bank.
- World Bank (2003). *Private sector development in the electric power sector: A joint OED/OEG/OEU review of the World Bank Group's assistance in the 1990s*. Washington DC: Operations Evaluation Department, World Bank.
- World Bank (2009). *Implementation completion and results report: fourth power project*. Available via: <http://documents.worldbank.org/curated/en/381721468109737499/pdf/ICR7600P00298411C0Disclosed05161091.pdf>, Accessed date: 6 November 2017.
- World Bank (2015). *The growth challenge: Can Ugandan cities get to work? Uganda economic update* (5th edition). Washington, D.C.: World Bank.
- World Bank (2016). *Private participation in infrastructure (PPI) database*. Available via: <https://ppi.worldbank.org/>, Accessed date: May 2013.
- World Bank (2017a). *Access to electricity (% of population) database*. Available via <https://data.worldbank.org/indicator/EG.ELC.ACCS.ZS>, Accessed date: 11 October 2017.
- World Bank (2017b). *Net official development assistance and official aid received (current US\$) database*. Available via <https://data.worldbank.org/indicator/DI.ODA.ALLD.CD?locations=UG>, Accessed date: 16 October 2017.
- World Bank and USAID (1994). *Submission and evaluation of proposals for private power generation projects in developing countries'. IEN occasional paper no. 2*. Washington, DC: World Bank.
- Yergin, D., & Stanislaw, J. (2002). *The commanding heights: The battle for the world economy*. New York: Simon & Schuster.